

Lecture 12: Public-Key Encryption Schemes

Notes by Yael Kalai

MIT - 6.5620

Lecture 12 (October 15, 2025)

Warning: This document is a rough draft, so it may contain bugs. Please feel free to email me with corrections.

Recap:

Last class we transitioned to public-key cryptography!

- Presented the Diffie-Hellman Key Agreement protocol.
- Defined public-key encryption scheme.
- Presented the El-Gamal encryption scheme.

Today

- Prove the security of the El-Gamal encryption scheme, and its variants.
- Define trapdoor permutations
- Construct a public-key encryption scheme from any trapdoor permutation.

The El-Gamal Encryption Scheme

- Let $G = G_\lambda$ be a group of prime order q of size roughly 2^λ , and let $g \in G$ be a generator. The message space $\mathcal{M}_\lambda$ is G .
- $\text{Gen}(1^\lambda)$:
 1. Choose at random $s \leftarrow \mathbb{Z}_q$.
 2. Set $\text{pk} = g^s$ and $\text{sk} = s$.
 3. Output (pk, sk) .
- $\text{Enc}(\text{pk}, m)$:
 1. Choose at random $r \leftarrow \mathbb{Z}_q$.
 2. Output $\text{ct} = (g^r, \text{pk}^r \cdot m)$.
- $\text{Dec}(\text{sk}, \text{ct})$:
 1. Parse $\text{ct} = (R, \text{ct}')$.

2. Output $m = \text{ct}' \cdot (R^{-\text{sk}})$.

Theorem 1. *This scheme is CPA secure under the DDH assumption.*

Recall that the DDH assumption w.r.t. a group G prime order q with generator $g \in G$, states that

$$(g^a, g^b, g^{a \cdot b}) \approx (g^a, g^b, g^c)$$

where $a, b, c \leftarrow \mathbb{Z}_q$.

Recall the definition of CPA security for public-key encryption:

Definition 2. A public-key encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ with message space $\mathcal{M} = \{\mathcal{M}_\lambda\}_{\lambda \in \mathbb{N}}$ is said to be CPA-secure if for every poly-size $\mathcal{A}$ there exists a negligible μ such that for every $\lambda \in \mathbb{N}$ the adversary $\mathcal{A}$ wins the game below with probability at most $\frac{1}{2} + \mu(\lambda)$:

1. The challenger samples $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$ and sends pk to $\mathcal{A}$.
2. $\mathcal{A}$ chooses $m_0, m_1 \in \mathcal{M}_\lambda$ and send these messages to the challenger.
3. The challenger samples a random bit $b \leftarrow \{0, 1\}$ and computes $\text{ct}_b \leftarrow \text{Enc}(\text{pk}, m_b)$. and sends ct_b to $\mathcal{A}$.
4. $\mathcal{A}$ outputs a bit b' (as a guess for b).

$\mathcal{A}$ wins if and only if $b' = b$.

Proof of Theorem 1 Fix any poly-size adversary $\mathcal{A}$ and suppose that $\mathcal{A}$ wins in the CPA security game (Definition 2) with non-negligible advantage $\epsilon : \mathbb{N} \rightarrow [0, 1]$. We construct a poly-size adversary $\mathcal{B}$ that breaks the DDH assumption. Given (g^a, g^b, g^c) , $\mathcal{B}$ emulates the CPA security game with $\mathcal{A}$, as follows:

1. Send $\text{pk} = (g, g^a)$ to $\mathcal{A}$.
2. Emulate $\mathcal{A}(\text{pk})$ to compute $m_0, m_1 \in G$.
3. Sample a random $b \leftarrow \{0, 1\}$, and send to $\mathcal{A}$ the ciphertext $(g^b, g^c \cdot m_b)$.
4. Compute the guess b' computed by $\mathcal{A}$.
5. If $b = b'$ then output 1 (indicating that the input (g^a, g^b, g^c) is a DDH tuple) and otherwise output 0.

Note that if (g^a, g^b, g^c) is a DDH tuple then this is a perfect simulation of the CPA game and hence, by our assumption

$$\Pr[\mathcal{B}(g^a, g^b, g^c) = 1 \mid (g^a, g^b, g^c) \in \text{DDH}] \geq \frac{1}{2} + \epsilon(\lambda)$$

On the other hand,

$$\Pr_{a,b,c \leftarrow [q]} [\mathcal{B}(g^a, g^b, g^c) = 1] = \frac{1}{2}$$

even if $\mathcal{B}$ is all powerful, since g^c is a perfect mask of m_b . □

Basically, the above proof shows that if we run the Diffie-Helman key agreement protocol polynomially many times $t = \text{poly}(\lambda)$, with the same first message g^a and with independent randomly chosen second messages $g^{b_1}, \dots, g^{b_t}$, then,

$$(g^a, g^{b_1}, \dots, g^{b_t}, g^{a \cdot b_1}, \dots, g^{a \cdot b_t}) \approx (g^a, g^{b_1}, \dots, g^{b_t}, g^{c_1}, \dots, g^{c_t})$$

where $c_1, \dots, c_t \leftarrow [q]$.

This can be proven via a standard hybrid argument (which if you squint, you can see that this is basically what is going on inside the CPA-security proof above). The point is that given

$$(g^a, g^{b_1}, \dots, g^{b_i}, g^{a \cdot b_1}, \dots, g^{a \cdot b_i})$$

we can simulate the rest of the elements

$$g^{b_{i+1}}, \dots, g^{b_t}, g^{a_{i+1} \cdot b_{i+1}}, \dots, g^{a \cdot b_t}$$

efficiently.

An Alternative Formulation of the El-Gamal Encryption Scheme

In the El-Gamal encryption scheme we presented above, the message space is G and we use g^{ab} as a one time pad by multiplying it with the message $m \in G$. Alternatively, we can think of the message space as being $\{0, 1\}^\lambda$ and defining the encryption algorithm to be:

$$\text{Enc}(\text{pk}, m) = (g^r, H(pk^r) \oplus m)$$

where $H : G \rightarrow \{0, 1\}^\ell$.

What properties do we need from H ? Well, it depends what it assumption we want to make about G . If we assume that the DDH assumption holds in the group G , then all we need is that $2^\ell \leq |G|$ and H can simply embed G in $\{0, 1\}^\ell$ by associating with each element $g \in G$, a lexicographic order $\ell(g) \in [|G|]$ and letting $H(g) = \ell(g) \bmod 2^\ell$.

However, another option is to rely on the weaker assumption that CDH holds in G , and rely on a stronger (cryptographic) assumption about H . For example, suppose that H “behaves like a truly random function.” Namely, suppose all the adversary can do is use H as a black-box, feed it with inputs and receive outputs. Moreover, suppose

that this black-box access is indistinguishable from black-box access to a truly random function. Then if the adversary cannot compute pk' then $H(pk')$ is indistinguishable from uniform, and hence can be safely used as a one-time pad.

Random Oracle Model

The random oracle model was introduced by Bellare and Rogaway [1]. The idea is to analyze the security of the cryptosystem assuming that the underlying hash function H is a truly random function to which the adversary has black-box access.

Note that this model is an *ideal* model, which is not consistent with reality! Hash functions are *not* random functions, and the adversary is given more than black-box access to the function, since it is given a succinct description of H that allows him to compute the function efficiently. Indeed, there are known counter examples, due to Canetti, Goldreich and Halevi [2], of schemes that are secure in the random oracle model and are insecure when the random oracle is instantiated with *any* hash function.

Nevertheless, the random oracle model has proven to be extremely useful! It is a great proxy for analyzing the security of cryptosystems, and so far we have not seen examples of real-world schemes that were proven secure in the random oracle model, and were broken in practice where the hash function was taken to be an off-the-shelf hash function, such as SHA256.

Constructing Public-Key Encryption from General Assumptions

So far we have seen how to construct a public-key encryption scheme from a very specific assumption, which may or may not be secure! Diffie and Hellman, in their seminal paper [3], proposed to use trapdoor functions to construct a public-key encryption.

Trapdoor One-Way Functions

Definition 3. A trapdoor function family consists of a PPT key generation algorithm Gen , that takes as input 1^λ and outputs a (public) hash key hk together with an associated (secret) trapdoor key td , and a family of functions $F_{hk} : \mathcal{D}_{hk} \rightarrow \mathcal{R}_{hk}$. The following properties are required to hold:

- **Efficient computable:** There is a poly-time algorithm $\mathcal{A}$ such that for every $(hk, td) \in \text{Gen}(1^\lambda)$ and for every $x \in \mathcal{D}_{hk}$, $\mathcal{A}(hk, x) = F_{hk}(x)$.

- **Efficiently sampleable:** There is a PPT algorithm $\mathcal{A}$ such that for any $(hk, td) \leftarrow \text{Gen}(1^\lambda)$, and any $x \in \mathcal{D}_{hk}$,

$$\Pr[\mathcal{A}(hk) = x] = \frac{1}{|\mathcal{D}_{hk}|}$$

(I.e., $\mathcal{A}$ outputs a uniformly random sample from the domain $\mathcal{D}_{hk}$)

- **Trapdoor invertible:** There is a poly-time inversion algorithm $\mathcal{I}$ such that for any $(hk, td) \in \text{Gen}(1^\lambda)$ and any $x \in \mathcal{D}_{hk}$

$$\mathcal{I}(td, F_{hk}(x)) = x$$

- **One-way:** For every polys-size $\mathcal{A}$ there exists a negligible function μ such that for every $\lambda \in \mathbb{N}$,

$$\Pr[\mathcal{A}(hk, F_{hk}(x)) = x] \leq \mu(\lambda)$$

where the probability is over $(hk, td) \leftarrow \text{Gen}(1^\lambda)$ and over $x \leftarrow \mathcal{D}_{hk}$.

Public-key Encryption from Trapdoor Functions

Intuitively, to build a public-key encryption scheme from a trapdoor function, we should take the public key to be hk and the secret key to be td . However, it is not clear what to do next. We will rely on the Goldreich-Levin theorem that there exists a hardcore predicate $\text{HCP}(x, r) = x \cdot r$ such that

$$(hk, F_{hk}(x), r, \text{HCP}(x, r)) \approx (hk, F_{hk}(x), r, U)$$

where $U \leftarrow \{0, 1\}$.

- **Gen:** Takes as input 1^λ and does the following:
 1. Sample $(hk, td) \leftarrow \text{Gen}_F$.
 2. Set $pk = hk$ and $sk = td$.
 3. Output (pk, sk) .
- **Enc:** takes as input the public key hk and a message $m \in \{0, 1\}^*$, and does the following:
 1. Sample a random $x \leftarrow \mathcal{D}_{hk}$.
 2. Compute $y = F_{hk}(x)$.
 3. Choose a random $r \leftarrow \{0, 1\}^\lambda$, and let $b = \text{HCP}(r, x)$.
 4. Output $(y, r, b \oplus m)$.
- **Dec:** On input a secret key td and a ciphertext $ct = (y, r, c)$, does the following:

1. Compute $x = \mathcal{I}(\text{td}, y)$.
2. Compute $b = \text{HCP}(r, x)$.
3. Output $m = b \oplus c$.

Theorem 4. *The public-key encryption scheme above is CPA-secure*

Proof. Suppose for the sake of contradiction that there exists a poly-size adversary $\mathcal{A}$ and a non-negligible ϵ such that $\mathcal{A}$ wins in the CPA game with probability ϵ . We construct a poly-size adversary that inverts the one-way function. To this end, it suffices to construct a poly-size $\mathcal{B}$ for which there exists a non-negligible ϵ such that for every $\lambda \in \mathbb{N}$,

$$\Pr[\mathcal{B}(\text{hk}, F_{\text{hk}}(x), r) = \text{HCP}(x, r)] \geq \epsilon(\lambda).$$

$\mathcal{B}(\text{hk}, F_{\text{hk}}(x), r)$ emulates the adversary $\mathcal{A}$ in the CPA game, as follows:

1. Send $\text{pk} = \text{hk}$ to $\mathcal{A}$.
2. Emulate $\mathcal{A}(\text{hk})$ to obtain $m_0, m_1 \in \{0, 1\}$.
3. Choose at random $b \leftarrow \{0, 1\}$.
4. For each $d \in \{0, 1\}$ (which should be thought of as a guess for $\text{HCP}(x, r)$),
 - (a) Let $\text{ct}_d = (F_{\text{hk}}(x), r, d \oplus m_b)$.
 - (b) Emulate $\mathcal{A}$ on input ct_d to obtain b'_d .
5. If $b'_0 = b'_1$ then output a random guess $b' \leftarrow b$.
6. Otherwise output d such that $b'_d = b$.

Note that for $d = \text{HCP}(x, r)$ it holds that

$$\Pr[b'_d = b] \geq \frac{1}{2} + \epsilon(\lambda)$$

whereas for a random $d \leftarrow \{0, 1\}$,

$$\Pr[b'_d = b] = \frac{1}{2}.$$

therefore $\mathcal{B}$ outputs $\text{HCP}(x, r)$ with a non-negligible advantage, as desired.

□

Examples of Trapdoor Functions

Rabin Function

$$F_N(x) = x^2 \bmod N$$

RSA Function

References

- [1] Mihir Bellare and Phillip Rogaway. Random oracles are practical: A paradigm for designing efficient protocols. In *Proceedings of the 1st ACM Conference on Computer and Communications Security (CCS '93)*, pages 62–73, Fairfax, Virginia, USA, 1993. ACM.
- [2] Ran Canetti, Oded Goldreich, and Shai Halevi. The random oracle methodology, revisited (preliminary version). In *Proceedings of the 30th Annual ACM Symposium on Theory of Computing (STOC '98)*, pages 209–218, Dallas, Texas, USA, 1998. ACM.
- [3] Whitfield Diffie and Martin E. Hellman. New directions in cryptography. *IEEE Transactions on Information Theory*, 22(6), 1976.